

STEPANOVKA, Russian Federation

WMO: 291490

Lat: 58.640N

Lon: 86.730E

Elev: 105

StdP: 100.07

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.8	(c) -37.9	(d) -43.9	(e) 0.1	(f) -40.8	(g) -40.5	(h) 0.1	(i) -37.4	(j) 8.2	(k) -13.2	(l) 7.2	(m) -10.1	(n) 0.9	(o) 20	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.4	18.5	27.6	17.4	25.9	17.0	20.2	26.3	19.1	24.8	18.2	23.6	2.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	22.3	17.1	12.4	21.3	16.1	11.6	20.7	58.6	26.2	55.1	25.0	52.0	23.5	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.4	(c) 5.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 7.6	(5) 6.4	(6) 5.4	(4) -43.3	(5) 21.8	(6) 2.1	(7) 0.9	(8) -44.8	(9) 22.5	(10) -46.1	(11) 23.0	(12) -47.3	(13) 23.6	(14) -48.8	(15) 24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-20.2	-16.9	-7.4	0.8	7.4	15.7	18.5	15.2	8.5	-0.1	-12.0	-16.9	(6)
(7)		DBStd	14.89	9.18	8.72	6.93	5.43	5.16	5.35	3.37	3.57	4.56	5.38	8.94	9.85	(7)
(8)		HDD10.0	4531	937	753	541	277	117	15	1	2	82	314	659	833	(8)
(9)		HDD18.3	6974	1196	986	799	526	340	111	39	108	296	572	909	1092	(9)
(10)		CDD10.0	688	0	0	0	1	36	186	264	164	36	1	0	0	(10)
(11)		CDD18.3	91	0	0	0	0	2	32	45	12	1	0	0	0	(11)
(12)		CDH23.3	1053	0	0	0	1	34	398	488	123	9	0	0	0	(12)
(13)		CDH26.7	258	0	0	0	0	4	115	122	17	1	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.1	2.2	2.6	2.8	2.8	2.4	2.0	1.9	2.0	2.4	2.6	2.2	(14)
(15)	Precipitation	PrecAvg	679	41	33	37	42	55	66	71	72	61	70	72	57	(15)
(16)		PrecMax	887	84	59	84	75	127	106	121	124	92	103	113	102	(16)
(17)		PrecMin	480	3	10	8	16	23	16	13	30	18	28	37	18	(17)
(18)		PrecStd	95	17	12	18	14	23	20	28	25	19	18	20	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	1.0	7.9	16.5	27.1	32.4	31.2	28.8	24.3	15.5	4.9	0.2	(19)
(20)			MCWB	-1.4	-0.6	2.3	7.9	14.5	18.3	20.6	19.3	16.1	8.7	3.3	-0.5	(20)
(21)		2%	DB	-3.0	-0.8	5.4	13.8	23.3	29.6	29.7	26.4	20.9	11.8	1.3	-1.0	(21)
(22)			MCWB	-3.7	-1.8	1.1	7.1	12.9	17.4	19.5	18.1	14.5	7.3	0.4	-1.5	(22)
(23)		5%	DB	-5.1	-2.7	3.5	11.2	19.7	27.6	28.0	24.5	18.0	9.5	0.1	-2.6	(23)
(24)			MCWB	-5.7	-3.6	0.0	5.2	11.3	16.7	17.9	17.3	13.2	6.5	-0.5	-3.1	(24)
(25)		10%	DB	-7.2	-4.9	1.7	8.8	16.5	25.5	26.2	22.5	15.4	7.3	-1.3	-4.6	(25)
(26)			MCWB	-7.7	-5.9	-0.5	3.8	10.0	16.3	17.5	16.0	11.6	5.0	-1.9	-5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-0.3	3.6	9.6	15.6	20.2	22.5	20.3	16.7	10.0	3.5	-0.2	(27)
(28)			MCDWB	-1.0	0.8	7.2	14.7	24.2	28.4	27.5	26.1	22.5	12.4	5.2	0.0	(28)
(29)		2%	WB	-3.6	-1.8	1.7	7.6	14.0	19.0	20.9	19.1	15.1	8.2	0.6	-1.6	(29)
(30)			MCDWB	-3.0	-0.8	4.1	12.5	21.0	26.1	26.9	24.2	20.3	11.8	1.2	-1.1	(30)
(31)		5%	WB	-5.7	-3.5	0.5	6.0	12.4	18.0	19.8	18.2	13.5	6.5	-0.5	-3.1	(31)
(32)			MCDWB	-5.2	-2.7	2.8	10.5	19.0	25.4	25.3	22.7	17.4	9.2	0.0	-2.6	(32)
(33)		10%	WB	-7.8	-5.9	-0.5	4.4	10.5	16.9	18.8	17.2	12.0	4.9	-1.9	-5.0	(33)
(34)			MCDWB	-7.2	-5.0	1.7	8.0	15.4	23.5	24.2	21.1	14.8	7.3	-1.3	-4.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	9.7	11.0	9.8	9.8	11.1	11.3	10.6	8.6	6.1	7.1	7.7	(35)
(36)			MCDBR	9.6	6.3	11.2	12.6	14.6	14.9	14.4	13.9	12.4	9.5	6.0	7.2	(36)
(37)			MCWBR	9.2	5.6	8.0	7.5	7.0	5.0	5.5	6.4	6.7	6.2	5.6	7.1	(37)
(38)		5% WB	MCDBR	9.7	6.3	8.5	11.3	12.9	12.5	12.0	11.8	11.6	9.1	6.2	7.1	(38)
(39)			MCWBR	9.3	5.7	6.4	7.6	6.9	4.8	5.8	6.1	6.7	6.4	6.0	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.315	0.356	0.378	0.396	0.399	0.444	0.407	0.365	0.336	0.295	0.276	(40)	
(41)		taud	1.976	1.980	1.949	2.075	2.257	2.340	2.192	2.286	2.356	2.261	2.035	1.925	(41)	
(42)		Ebn at Noon	493	666	753	817	833	837	782	787	764	658	468	350	(42)	
(43)		Edh at Noon	57	93	129	136	122	114	130	110	88	71	53	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.58	1.54	2.99	4.43	4.86	5.70	5.25	4.00	2.41	1.18	0.55	0.32	(44)	
(45)		RadStd	0.07	0.13	0.20	0.34	0.50	0.57	0.43	0.31	0.34	0.14	0.12	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (2 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(46) Station Only	(47) Regional (2 neighbors)	(46) N/A	(47) N/A	(46) N/A	(47) N/A	(46) N/A	(47) N/A	(46) N/A	(47) N/A	(46) N/A

Nomenclature: See separate page